EUGENE OWINO

Staff Software Engineer | AI Evaluation | Fintech Systems

Toronto, ON, Canada | [Phone] | eugeneowino.dev@gmail.com | [LinkedIn URL] | [GitHub URL]

PROFESSIONAL SUMMARY

Staff software engineer with 7+ years of experience building fintech platforms and AI evaluation systems for regulated financial services. Leads a team of 8 engineers building a production LLM/RAG evaluation platform supporting KYC, AML, fraud, and compliance workflows, with CI/CD quality gates that cut post-deployment AI incidents by 35%. Part-time PhD in Computer Science (University of Waterloo) focused on reliable evaluation of foundation models in finance. Builds in Python, Go, Java, TypeScript, and SQL on AWS, GCP, and Azure.

CORE SKILLS

Languages: Python, Go, Java, TypeScript, JavaScript, SQL, C++, Bash, Rust (working proficiency)

AI Evaluation: LLM and RAG evaluation, RLHF preference ranking, benchmark design, golden datasets, rubric-based scoring, LLM-as-judge calibration, adversarial testing, red teaming, hallucination detection, drift monitoring, statistical significance testing, human-in-the-loop evaluation, model cards

ML and Data: PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, scikit-learn, pandas, MLflow, Weights & Biases, Ray, embeddings, vector search, Airflow, dbt

Fintech Domain: Payments, transaction monitoring, fraud detection, KYC/AML, ledger reconciliation, risk scoring, regulatory reporting, audit trails

Infrastructure: AWS, GCP, Azure, Kubernetes, Docker, Terraform, GitHub Actions, Kafka, PostgreSQL, Redis, REST, gRPC, Prometheus, Grafana, OpenTelemetry

PROFESSIONAL EXPERIENCE

Staff Software Engineer, AI Evaluation & Fintech Platforms **Feb 2024 – Present**

Northwind Financial Technologies | Toronto, ON

Tech: Python, Go, Java, TypeScript, SQL, FastAPI, LangChain, Kafka, AWS, Kubernetes, Terraform, MLflow

- Lead a team of 8 engineers building a production AI evaluation platform for LLM and RAG systems used in KYC document review, AML case summarization, fraud operations, payment support, and compliance Q&A.
- Reduced post-deployment AI incidents by 35% by adding CI/CD quality gates (GitHub Actions, Docker, Kubernetes, Terraform) that automatically block unsafe model and prompt changes.
- Increased answer faithfulness by 34% by rebuilding retrieval over regulatory policies, fraud playbooks, and support documents using embeddings, reranking, metadata filtering, and citation tracking.
- Architected automated evaluation pipelines (Python, FastAPI, Pytest, MLflow) with statistical significance testing to regression-test every prompt version, retrieval configuration, model checkpoint, and safety policy.
- Engineered Go services that ingest and score thousands of evaluation events per minute, integrated with core Java and SQL fintech systems.
- Deployed PII redaction, prompt-injection classification, policy-violation monitoring, and audit logging, and delivered TypeScript/React review dashboards for risk and compliance teams.
- Translated regulatory requirements into measurable evaluation criteria with product, risk, compliance, and data science teams; mentored engineers on evaluation-driven development.

Senior AI Model Evaluation Engineer **Jan 2022 – Jan 2024**

Meridian Payments | Toronto, ON

Tech: Python, Java, TypeScript, SQL, PyTorch, Hugging Face, LangChain, Kafka, PostgreSQL, MLflow

- Improved LLM-judge agreement with human analysts on high-risk case categorization by 28% by building judge calibration pipelines.
- Found and mitigated 15 high-severity issues before launch through adversarial testing for prompt injection, data leakage, policy bypass, and unsafe financial advice.
- Cut false escalation rates by 22% by evaluating model outputs against known fraud typologies, payment policies, and support workflows.
- Developed benchmark suites and golden datasets measuring accuracy, calibration, robustness, explainability, and policy compliance, integrated with real-time fraud systems via Java microservices and Kafka.
- Launched a TypeScript/React fraud-explanation quality dashboard combining analyst feedback, case outcomes, drift metrics, and searchable audit logs for compliance review.

- Automated CI regression testing for prompt, retrieval, and model upgrades; authored model cards and risk reports for compliance and executive stakeholders.

Graduate Research Assistant, Machine Learning for Financial Systems **Sep 2020 – Dec 2021**
University of Toronto | Toronto, ON
Tech: Python, C++, SQL, PyTorch, scikit-learn, pandas

- Built Python pipelines for evaluating fraud-detection and financial-risk models, covering feature engineering, benchmarking, statistical testing, and reproducibility.
- Wrote performance-critical C++ components for processing large transaction datasets in low-latency experiments.
- Designed SQL extraction and validation pipelines that improved dataset consistency and experiment readiness.

Software Engineer, Payments & ML Infrastructure **Jan 2019 – Aug 2020**
Lumen Ledger Labs | Toronto, ON
Tech: Java, Spring Boot, Python, TypeScript, SQL, PostgreSQL, Docker

- Developed core Java/Spring Boot payment services connecting ledgers, transaction APIs, and analytics for authorization, reconciliation, and risk scoring.
- Built Python model-serving and feature pipelines that deployed fraud and risk models into live payment workflows.
- Created an AI audit trail tracing model-assisted reconciliation decisions back to source transactions, features, and analyst actions.
- Designed SQL data models and reporting for reconciliation and audit, and shipped TypeScript operator dashboards for payment failures and case queues.

ADDITIONAL EXPERIENCE

AI Model Evaluation Specialist (Part-time Contract) **Oct 2024 – Present**
TELUS Digital, Alignerr, Outlier, Handshake AI | Remote
Tech: LLM evaluation platforms, rubric-based scoring tools, Python, SQL

- Evaluate LLM responses for accuracy, reasoning quality, instruction following, safety, and helpfulness against detailed client rubrics, with written rationales for each rating.
- Rank model outputs to produce RLHF preference data, and write adversarial prompts to expose hallucinations, reasoning failures, and policy violations.
- Review model-generated Python and SQL code and technical explanations for correctness and quality.

SELECTED PROJECT

FinEval-Bench: Financial AI Evaluation Benchmark
University of Waterloo & Northwind Financial Technologies

- Designed a modular benchmark evaluating LLM and RAG systems on financial reasoning, compliance Q&A, transaction explanation, adversarial robustness, and calibration.
- Built dataset versioning, evaluation harnesses, and leaderboard automation to compare models on accuracy, safety, latency, cost, and explainability.

EDUCATION

PhD, Computer Science (Part-time), Machine Learning / Fintech Systems **2020 – 2024**
University of Waterloo | Waterloo, ON
Dissertation: Reliable Evaluation of Foundation Models in Regulated Financial Systems

MSc, Computer Science | University of Toronto, Toronto, ON **2018 – 2020**

BSc, Computer Science | Stanford University, Stanford, CA, USA **2014 – 2018**

CERTIFICATIONS

- Databricks Certified Generative AI Engineer Associate | Databricks | Issued Jan 2025
- HashiCorp Certified: Terraform Associate | HashiCorp | Issued Jul 2024
- Certified Kubernetes Application Developer (CKAD) | The Linux Foundation / CNCF | Issued Apr 2024
- AWS Certified Machine Learning – Specialty | Amazon Web Services | Issued Mar 2024
- Google Cloud Professional Machine Learning Engineer | Google Cloud | Issued Feb 2024
- Microsoft Certified: Azure AI Engineer Associate | Microsoft | Issued Nov 2023